Circuits Updates - April 2024

We report a number of developing ideas on the Anthropic interpretability team, which might be of interest to researchers working actively in this space. Some of these are emerging strands of research where we expect to publish more on in the coming months. Others are minor points we wish to share, since we're unlikely to ever w paper about them.

We'd ask you to treat these results like those of a colleague sharing some thoughts or preliminary experiments few minutes at a lab meeting, rather than a mature paper.

New Posts

- Open Roles In Interpretability
- Scaling Laws for Dictionary Learning
- Update on how we train SAEs
- How Strongly do Dictionary Learning Features Influence Model Behavior?
- Interpretability Architectures Project
- Caloric and the Utility of Incorrect Theories
- Open Problem: Attribution Dictionary Learning
- Isolating Circuits Paths of Different Lengths
- Research By Other Groups

Open Roles in Mechanistic Interpretability

Chris Olah, Shan Carter, Adam Jermyn, Josh Batson, Tom Henighan

Mechanistic interpretability is a small field, although growing quite quickly. We estimate there are perhaps 50 fu time positions focused on this topic. The Anthropic interpretability team is now 17 people, so we represent a significant fraction of these positions. As a result, we felt that providing some visibility into our hiring plans mig valuable for people considering careers in this space. (However, please note that while these are our present expectations, they are subject to change.)

Over the course of 2023 we hired 10 people. We've continued hiring in 2024, and expect to continue growing t team substantially, both this year and into 2025. We expect this to involve a few different roles:

- **Managers** - We see this as the most important role that we're hiring for right now.
    - Our growth is likely to be bottlenecked on management capacity, and finding the right fit for th team could make a huge difference to our long-term success.
    - Filling this role has been challenging because we're looking for someone with experience in a research or engineering environment, who is excited about and experienced with people and pr management, and who is enthusiastic about our research agenda and mission.

- **Research Scientists** - We're looking for strong scientists, not necessarily experienced machine learning researchers. Most of our team comes from other backgrounds (astrophysics, condensed matter, mather neuroscience). We do want to see evidence of engagement with mechanistic interpretability, as well as sufficient coding ability to implement and run ambitious experiments.

- **Research Engineers** - We're looking for strong software engineers. Experience with machine learning i plus, but not necessary – we're excited to consider strong software engineers who want to grow into research. Our team has a track record of having several such people (eg. Nelson Elhage, Tristan Hume) j and quickly growing to perform state of the art research. Comfort with linear algebra, multivariate calcu and basic engagement with machine learning and mechanistic interpretability is a strong plus if you do have an ML background.

A few notes:

- Internally, we don't really distinguish between research engineers and scientists. All members of our tea both research and engineering. However, people are often stronger at one than the other, and we try to a mixture of these strengths for team balance.
- If you are interested in our new interpretability architectures project (see below), please apply to the res scientists or research engineer role and mention this in your application.

If you're excited about our work and think you might be a fit for one of these roles, please apply!

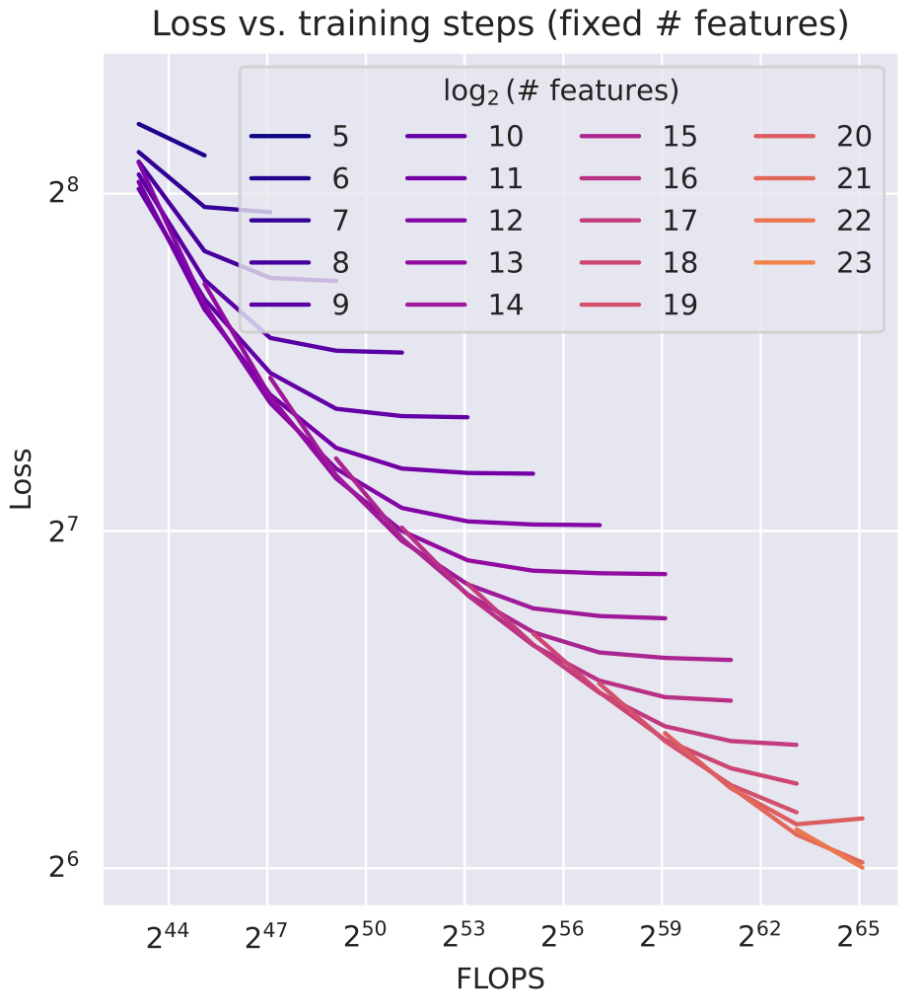
Scaling Laws for Dictionary Learning

Jack Lindsey, Tom Conerly, Adly Templeton, Jonathan Marcus, Tom Henighan

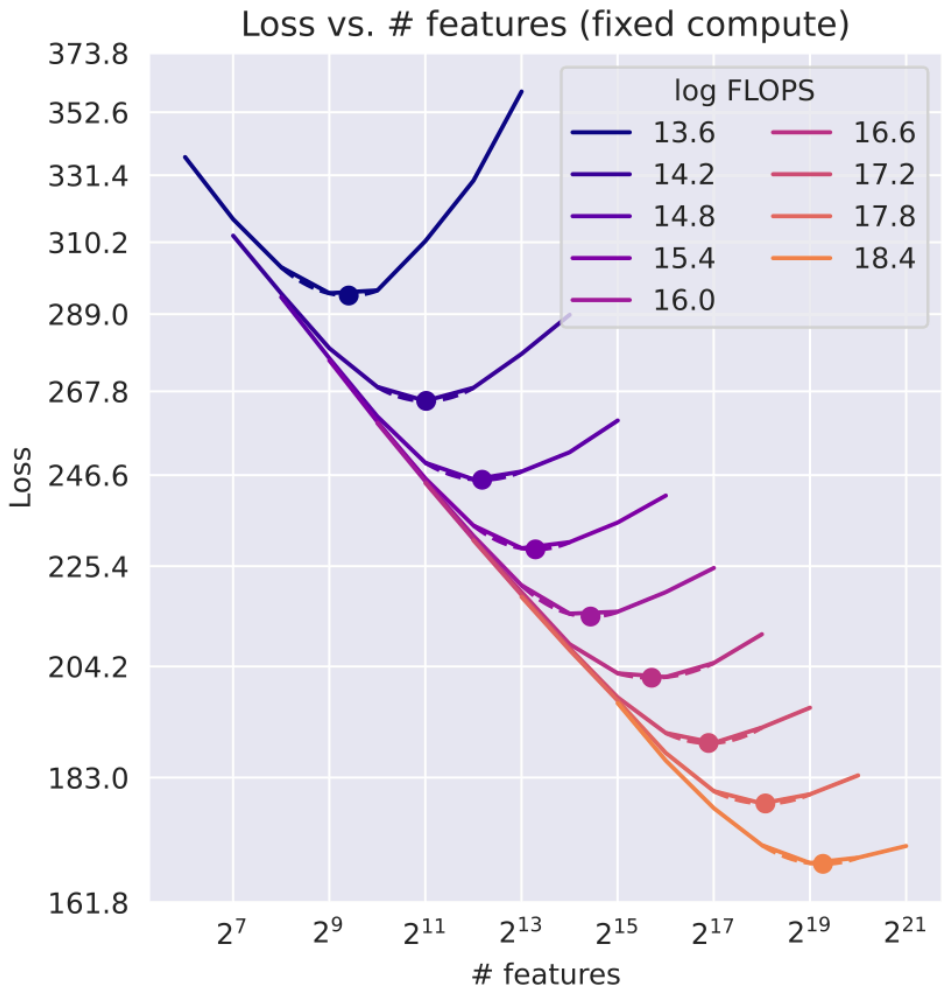
Training sparse autoencoders (SAEs) for dictionary learning on larger models can be computationally intensive. important to understand (1) the extent to which using additional compute improves dictionary learning results, (2) how that compute should be allocated. Here we analyze these questions in depth. As a case study, we cons SAEs trained on the residual stream following the third layer of a four-layer transformer.

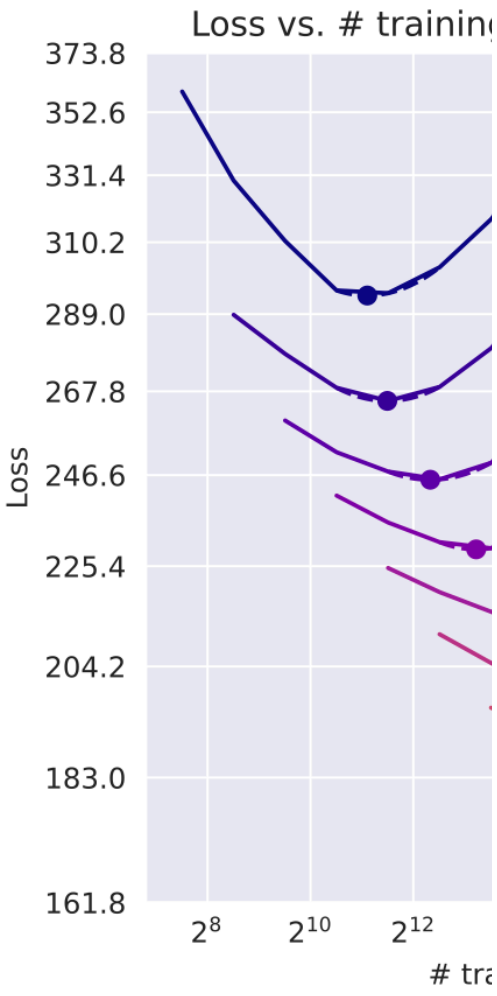
Though we lack a gold-standard method of assessing the quality of a dictionary learning run, we have found th loss function used during training – a weighted combination of reconstruction mean-squared error (MSE) and penalty on feature activations – is a useful proxy. Unless otherwise indicated, we use $\mathrm{MSE} + 5\mathrm{L1}$ as our loss function during training and subsequent analysis. However, we find qualitatively similar results when we use oth linear combinations of MSE and L1, or when we track linear combinations of MSE and the L0 "norm" of feature activations. We note that losses with different L1 coefficients are not comparable – ultimately, we select the L1 coefficients that produce the most useful features for downstream interpretability analyses.

Once we have chosen a loss function of interest, it allows us to treat dictionary learning as a standard machine learning problem, to which we can apply the "scaling laws" framework for hyperparameter optimization (*see e.g.* Kaplan *et al.* 2020, Hoffman *et al.* 2022). In an SAE, compute usage primarily depends on two key hyperparameters, the number of features being learned, and the number of steps used to train the autoencode compute (in FLOPS) scales with the product of these parameters, if the input dimension and other hyperparame are held constant. We conducted a thorough sweep over these parameters, fixing the values of other hyperparameters (learning rate, batch size, optimization protocol, etc.).



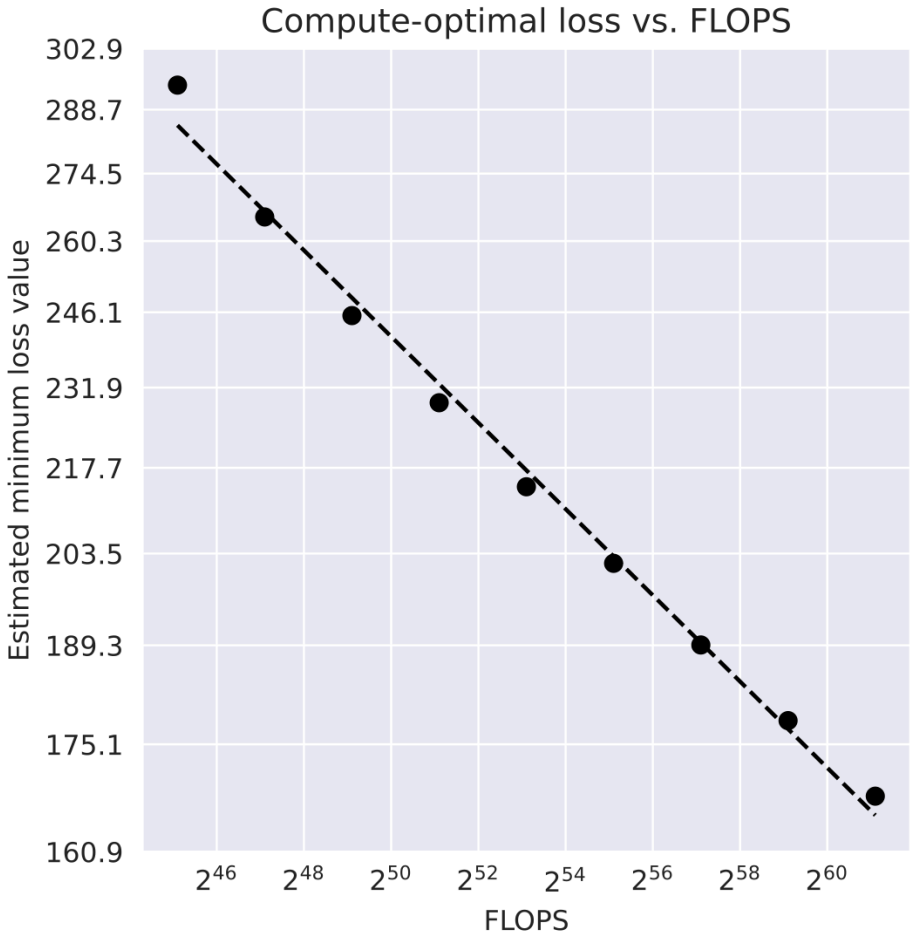
We are especially interested in keeping track of the compute-optimal values of the loss function and parameter interest; that is, the lowest loss that can be achieved using a given number of FLOPS, and the number of trainin steps / features that achieve this minimum.



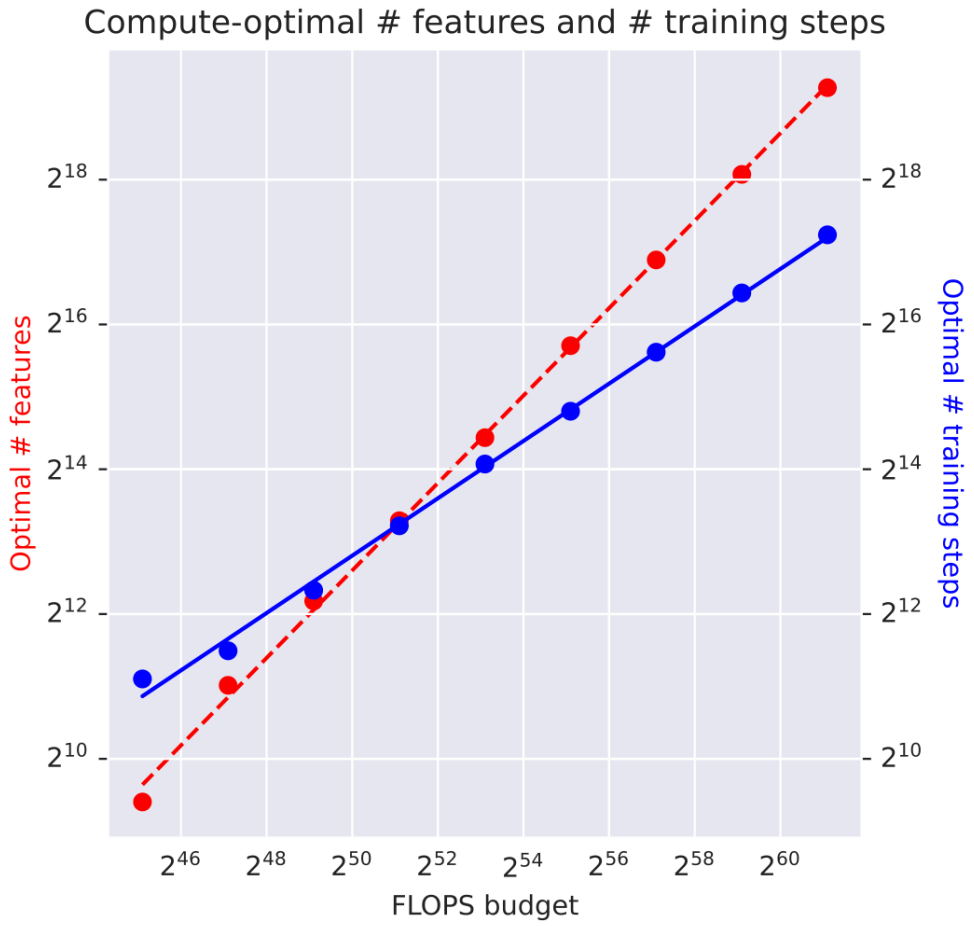


We have made the following observations:

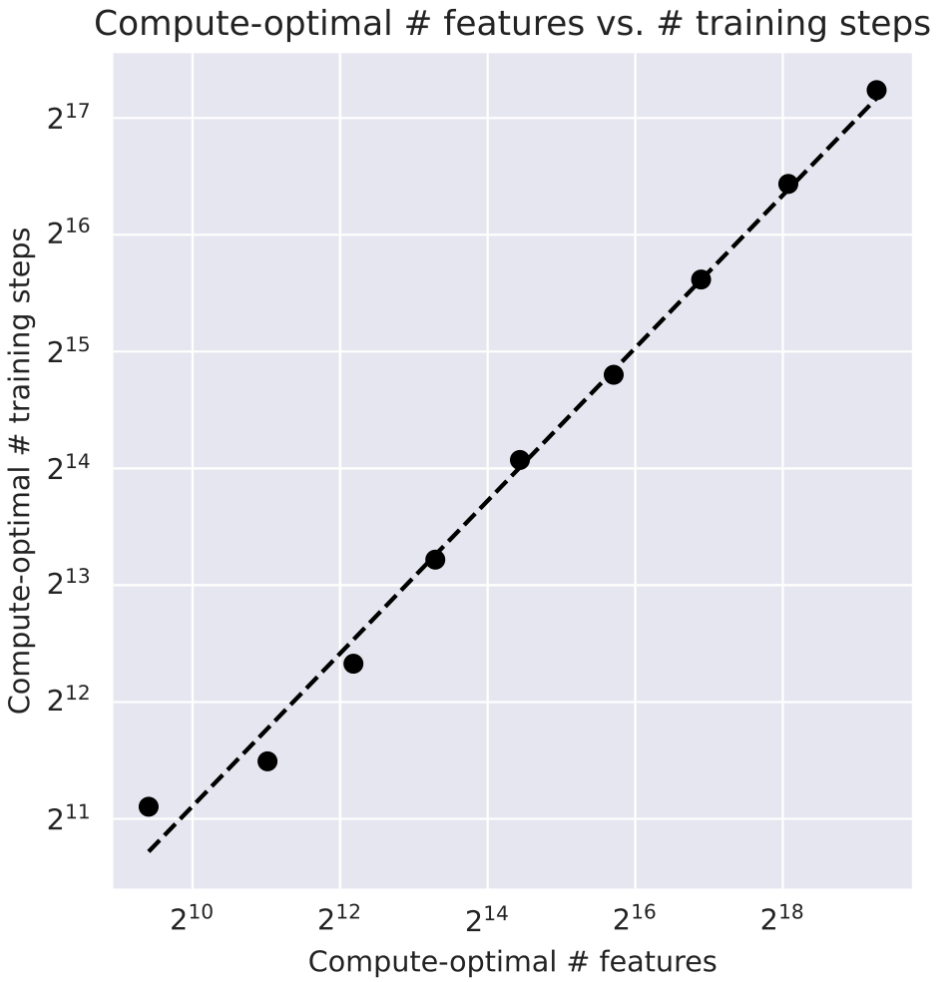
- Over the ranges we tested, loss functions decrease approximately according to a power law with respec FLOPS, given the compute-optimal choice of training steps and number of features.



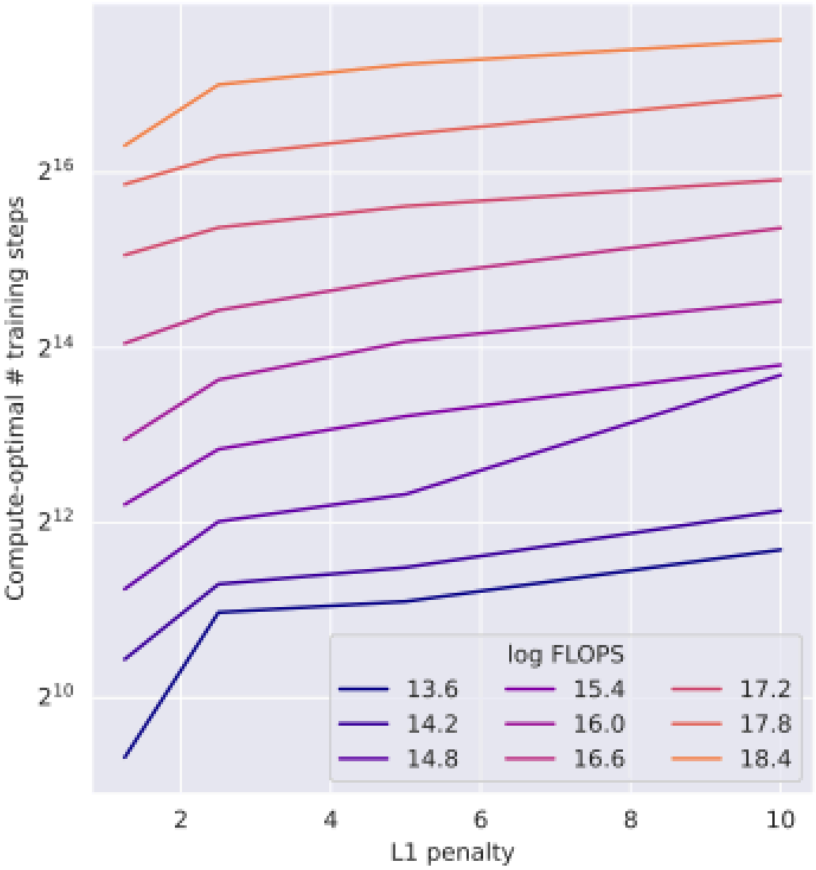
- As the compute budget increases, the optimal allocations of FLOPS to training steps and number of fea both scale approximately as power laws.

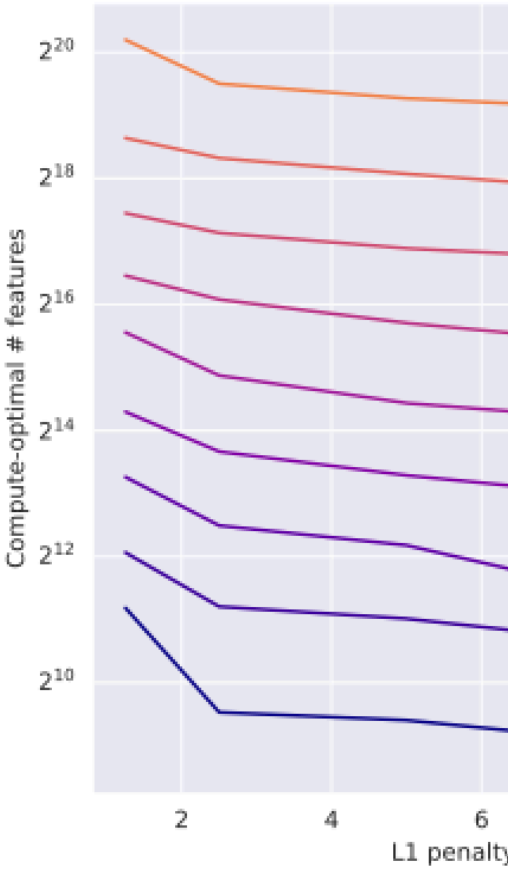


- In general, the optimal number of features appears to scale somewhat more quickly than the optimal n of training steps. For instance, as the number of features increases, the corresponding compute-optima number of training steps scales as a function of the number of features with an exponent between 0.5 a The specific parameters of the scaling trends vary between different loss functions.

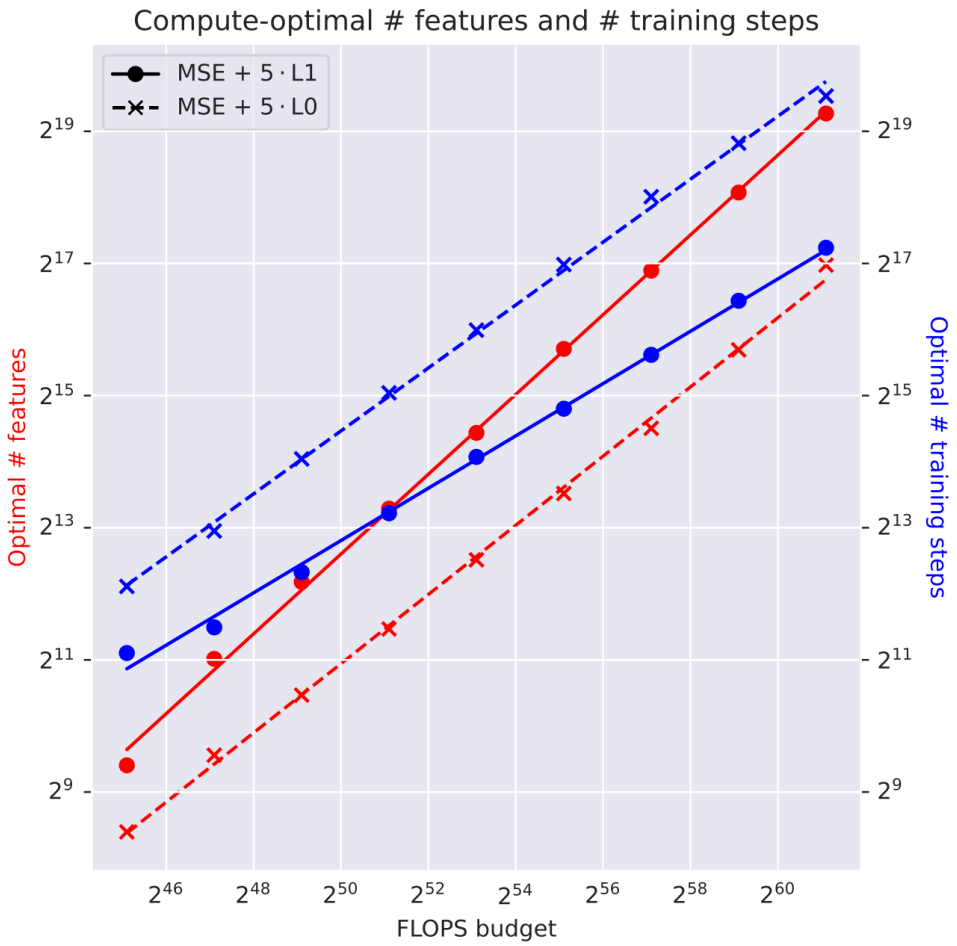


- Emphasizing the sparsity penalty more in the loss function (i.e. increasing the L1 coefficient) leads to a l number of training steps being compute-optimal.The compute-optimal number of training steps increa for loss functions that place a greater relative emphasis on the sparsity penalty. This suggests that reconstruction loss is optimized more quickly than sparsity over the course of SAE training, which we ha observed empirically.





- In these experiments, we also investigate the parameters needed to optimize for L0-based loss function linear combinations of MSE and L0 "norm" of feature activations). Since these cannot be directly optim with gradient descent, we instead sweep over a range of L1 penalty coefficients during training and sele value that minimizes the L0-based loss function. We find that minimizing L0-based loss functions requir greater number of training steps for a given number of features, compared to minimizing L1-based loss functions. Though we have not precisely characterized the source of this difference, we suspect it arises because additional training steps allow the SAE to fully zero out small feature activations.



The details of these trends are likely to vary depending on the underlying model, the layer of the model being probed, and other optimization details. Optimizing other hyperparameters (such as learning rate) jointly with training steps and number of features may influence the scaling trends. However, we expect many of these qualitative trends to be broadly applicable. We suggest that conducting similar analyses will be useful to other groups working with SAEs, particularly as computational cost increases. Extrapolating trends inferred from smal experiments enables more informed choices of hyperparameters for resource-intensive dictionary learning runs are also careful to note that qualitative inspection of SAE features remains important, as the relationship betwe SAE loss and qualitative usefulness of SAE features is imperfect and may break down at sufficient scale.

Update on how we train SAEs

Tom Conerly, Adly Templeton, Trenton Bricken, Jonathan Marcus, Tom Henighan

We've made improvements to how we train SAEs since Towards Monosemanticity with the goal of lowering the loss. While the new setup is a significant improvement over what we published in Towards Monosemanticity we believe there are further improvements to be made. We haven't ablated every decision so it's likely some simplifications could be made. This work was explicitly focused on lowering loss and didn't grapple with loss nc being the ultimate objective we care about. Here's a summary of our current SAE training setup:

Let n be the input and output dimension and m be the autoencoder hidden layer dimension. Let s be the size dataset. Given encoder weights $W_e \in \mathbb{R}^{m \times n}$, decoder weights $W_d \in \mathbb{R}^{n \times m}$, and biases $\mathbf{b}_e \in \mathbb{R}^m$, $\mathbf{b}_d \in \mathbb{R}^n$, t operations and loss function over a dataset $X \in \mathbb{R}^{s,n}$ are:

$$
\begin{aligned}
\mathbf{f}(x) &= \mathrm{ReLU}(W_e \mathbf{x} + \mathbf{b}_e) \\
\hat{\mathbf{x}} &= W_d \mathbf{f}(x) + \mathbf{b}_d \\
\mathcal{L} &= \frac{1}{|X|} \sum_{\mathbf{x} \in X} ||\mathbf{x} - \hat{\mathbf{x}}||_2^2 + \lambda \sum_i |\mathbf{f}_i(x)| ||W_{d,i}||_2
\end{aligned}
$$

Note that the columns of W_d have an unconstrained L2 norm (in Towards Monosemanticity they were constrai norm one) and the sparsity penalty (second term) has been changed to include the L2 norm of the columns of We believe this was the most important change we made from Towards Monosemanticity.

$\mathbf{b}_e$ and $\mathbf{b}_d$ are initialized to all zeros. The elements of W_d are initialized such that the columns point in random directions and have fixed L2 norm of 0.05 to 1 (set in an unprincipled way based on n and m, 0.1 is likely reaso in most cases). W_e is initialized to W_d^T.

The rows of the dataset X are shuffled. The dataset is scaled by a single constant such that $\mathbb{E}_{x \in X}[||x||_2] = \sqrt{}$ goal of this change is for the same value of λ to mean the same thing across datasets generated by different si transformers.

During training we use Adam optimizer beta1=0.9, beta2=0.999 and no weight decay. Our learning rate varies l on scaling laws, but 5e-5 is a reasonable default. The learning rate is decayed linearly to zero over the last 20% training. We vary training steps based on scaling laws, but 200k is a reasonable default. We use batch size 2048 4096 which we believe to be under the critical batch size. The gradient norm is clipped to 1 (using clip_grad_no We vary λ during training, it is initially 0 and linearly increases to its final value over the first 5% of training step reasonable default for λ is 5.

We do not use resampling or ghost grads because less than 1% of our features are dead at the end of training means not activating for 10 million samples). We don't do any fine tuning after training.

Conceptually a feature's activation is now $\mathbf{f}_i ||W_{d,i}||_2$ instead of $\mathbf{f}_i$. To simplify our analysis code we construct a model which makes identical predictions but has an L2 norm of 1 on the columns of W_d. We do this by $W'_e = W_e ||W_d||_2$, $b'_e = b_e ||W_d||_2$, $W'_d = \frac{W_d}{||W_d||_2}$ and $b'_d = b_d$.

Potential areas for improvement

Our initialization likely needs improvement. As we increase m the reconstruction loss at initialization increases. may cause problems for sufficiently large m. Potentially with improved initialization we could remove gradient clipping.

We haven't seen improvements in loss from resampling or ghost grads, but it's possible resampling "low value" features would improve loss.

It's plausible some sort of post training (for example Addressing Feature Suppression in SAEs) would be helpful

Improving shrinkage is an area for improvement.

There are likely other areas for improvement we don't know about.

Results

Given a fixed dataset X as we increase m the loss consistently decreases. We've been able to increase m to sir digit millions without issues. This holds across a variety of transformer sizes and mlp activations or the residual stream. Our setup from Towards Monosemanticity would frequently have higher loss, many dead features, or m nearly identical features when run with large values of m.

We make changes to our training setup by looking at loss across a variety of values of λ, m, transformer sizes, mlp or residual stream runs. We're generally excited by a change that consistently decreases loss by at least 1% change with roughly equal loss that simplifies our training setup. With our setup, comparing runs on (L0, MSE) % of MLP loss recovered) requires care because L0 can be unstable. For example we've had cases where training twice as long with half the number of features leads to a <1% change in MSE and L1 but a 30% decrease in L0.

Here are some results from small models. All runs have 131,072 features, 200k train steps, batch size 4096. Note L1 of f depends on our specific normalization of activations.

Type of Run	Lambda	L0(f)	L1(f)	Normalized MSE	Frac Cross Entropy Loss Recovered
1L MLP	2	99.62386	17.22560	0.03054	0.98305
1L MLP	5	38.68729	11.59591	0.06609	0.96398
1L MLP	10	20.06870	7.12194	0.13120	0.91426
4L MLP (layer 2)	2	264.02930	95.03488	0.06824	0.96824
4L MLP (layer 2)	5	69.92758	56.92384	0.12546	0.92904
4L MLP (layer 2)	10	26.48456	39.42661	0.18485	0.88438
4L Residual Stream (layer 2)	2	81.58595	30.37323	0.09543	0.9572
4L Residual Stream (layer 2)	5	33.23121	19.12259	0.16295	0.90443
4L Residual Stream (layer 2)	10	8.71466	12.53889	0.25455	0.83883

How Strongly do Dictionary Learning Features Influence Model Behavior?

Jack Lindsey

Features uncovered by sparse autoencoders are optimized to reconstruct model activity while remaining sparse active. Our team and others have observed that these features often appear to encode specific, interpretable concepts. However, a potential concern about using these features for an interpretability agenda is that, despite semantic significance to humans, these features may not capture the abstractions that the model uses for its computation. We have conducted preliminary experiments that suggest that models do in fact "listen" to featur values significantly more than would be expected by chance.

Our experiment works as follows: We train a sparse autoencoder (SAE) on the residual stream following the thir layer of a trained four-layer transformer. For each SAE feature, we take a representative sample of datapoints fc which that feature has a nonzero activation, scale the value of that activation by a factor of either 0 ("feature ablation") and 2 ("feature doubling"), and propagate the updated value through to the model according to the procedure as in Towards Monosemanticity (the case of scaling factor equal to zero corresponds to a feature abl We compute the average increase in the model's loss following this procedure.

Our goal is to determine whether the loss increase from rescaling feature activation values is especially high compared to other model perturbations with similar statistics. If so, it would provide evidence that feature direc exert "special" influence on downstream computation in the model. To this end, we compare feature rescaling t several controls:

- Apply a random perturbation to the residual stream activity at the same layer, matching the magnitude random perturbation to the magnitude of the perturbation induced by the feature rescaling ("random perturbation, model activations" in the figure).

This control is meant to test whether feature ablations are more significant than random perturbations. Arguab is a weak baseline, as the variance of model activations is likely not isotropic, and thus some dimensions of resi stream activity may be less consequential to model behavior. SAE feature vectors are trained to reconstruct mo activity, and as a result probably concentrate in more important dimensions. Thus, as a stronger baseline, we tri following:

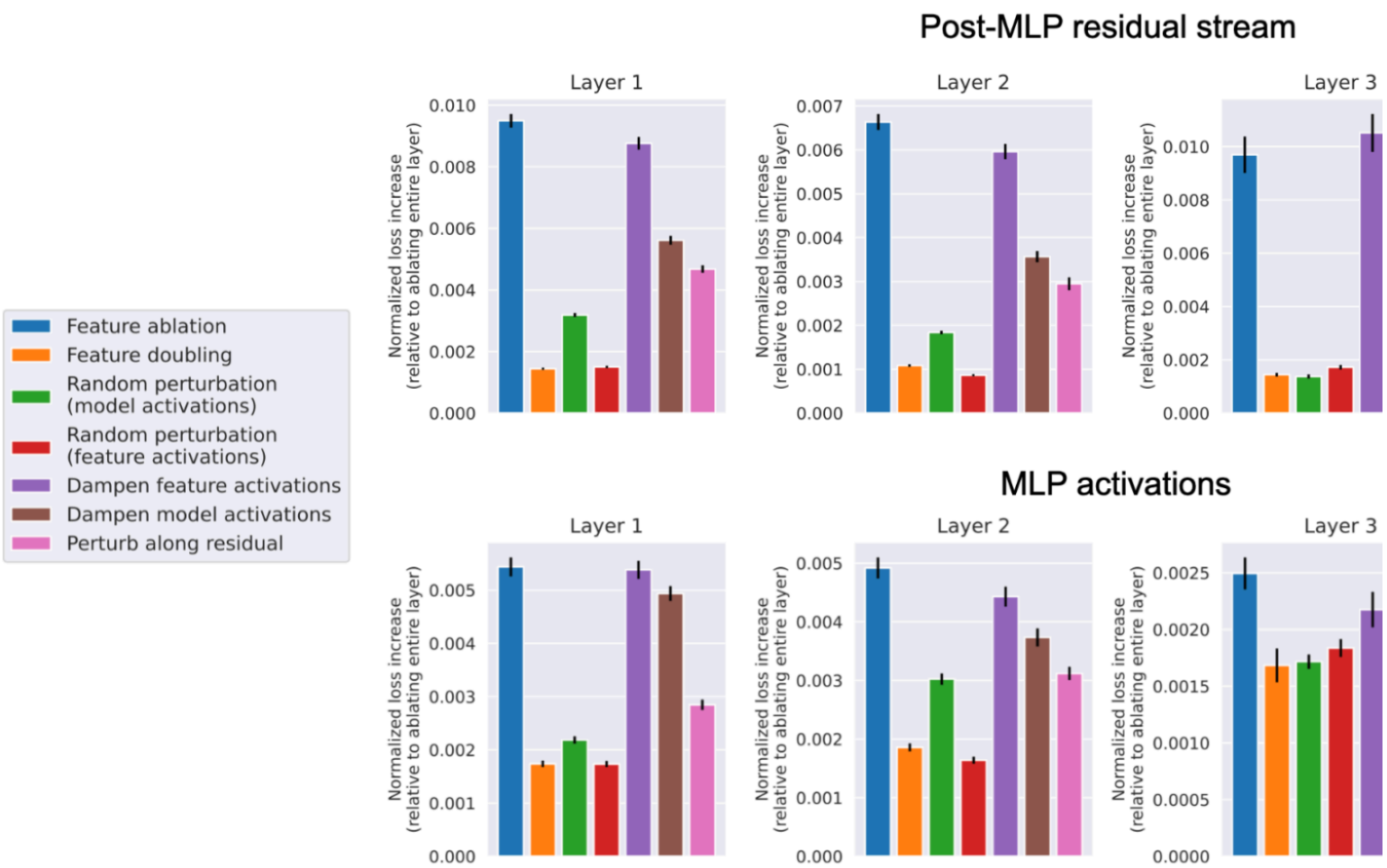
- Apply a random perturbation of the same magnitude as the feature rescaling on the feature activations vector ("random perturbation, feature activations" in the figure), and propagate that perturbation throu the model according to the same procedure used for the feature rescalings.

These experiments revealed several interesting findings:

- We found that feature ablations have significantly greater impact on model performance than either of controls.
- Interestingly, *ablating* the feature activation has a substantially greater effect on model performance tha *amplifying* the feature by the same amount, suggesting that the influence of features on model outputs (partially) saturate at higher feature activations.

We also compared the effect of feature perturbations to other, more structured forms of perturbations. In all ca
we match the magnitude of the perturbation in model activation space to be equal to that of the correspondin
feature ablation.

- Controlling for perturbation magnitude, applying perturbations in a direction antiparallel to the SAE reconstructions ("dampen feature activations" in the figure) – equivalent to "spreading out" a feature ab across all features, in proportion to their activity – produces similar effects as single-feature ablations.
- Controlling for magnitude, perturbations antiparallel to the model activity ("dampen model activations" figure) have less impact than feature dampening. Note that dampening model activations is different th dampening feature activations, as model activations include two components – the bias term of the SAE the error vector left unexplained by the SAE – that are not affected by feature dampening. This result indicates that the feature dampening effect is not explainable simply due to its effect on activation norr fact, in later layers, the effect of dampening model activations almost vanishes.
- Consistent with the results of Gurnee (discussed in detail elsewhere in this update), magnitude-controll perturbations along the reconstruction error direction -(x - SAE(x)) ("perturb along residual" in the figur also substantially more impactful than random perturbations, though less impactful than feature ablatic While not conclusive, these results suggest that Gurnee's findings are consistent with a model in which reconstruction errors lie primarily along feature directions, as this would explain their greater-than-rand impact on model outputs.
- The outsized impact of feature ablations is more pronounced for residual stream features than MLP laye features. This suggests that residual stream and MLP features may play different functional roles in the network, though our understanding of this result is limited.



In this figure, results are averaged over contexts, tokens, and features, and error bars indicate standard error of mean over features.

These results are preliminary, but generally support the idea that feature directions uncovered by SAEs are high leverage “levers” for influencing model outputs.

Interpretability Architectures Project

Chris Olah, Adam Jermyn

From time to time, we've noticed aspects of transformer architecture that make our lives on interpretability mo difficult. For instance, layernorm makes circuit analysis and attribution more difficult, and we've invested signific effort in trying to get rid of it over the years. Similarly, SoLU was an attempt at making models more interpretab although we ultimately believe it wasn't the right approach to that specific problem.

We believe it's possible that investing in model architecture now may save a lot of interpretability effort in the f For this reason, we're starting an experimental working group to explore more interpretable architectures. This working group will investigate architectural decisions that might make interpretability easier, and will collaborat the Pretraining team to support their implementation. For now, this working group will be smaller than the mai interpretability teams (dictionary learning, attention, and circuits). This working group will be embedded in both Interpretability and Pretraining, and members will sometimes contribute to projects on both of these broader te

If you're interested in this new working group, please apply to join our team and indicate interest in working on interpretable architectures (see above).

Caloric and the Utility of Incorrect Theories

Tom Henighan

"Caloric theory" is an outdated theory of heat which can be summarized as follows:

> *There is a massless, self-repelling substance called "caloric" which increases the temperature of whatever ma inhabits.*

It's easy to scoff at this theory and regard it as silly with the benefit of hindsight. But in fact, this theory could e many, if not most, thermodynamic measurements that were available at the time. Consider the phenomenon o flow as a concrete example. Under caloric theory, a hot (high temperature) object contains a lot of caloric. That caloric is self-repelling, but cannot spread out more because it's constrained by the boundaries of the hot obje we put this hot object into contact with a cold object, the caloric spreads into the cold object. The hot object c and the cold object heats as caloric flows from the former to the latter.

I now invite the reader to put themselves in the shoes of a 17th century scientist and design an experiment whi disprove caloric theory. Further, imagine how much more challenging this would be if you didn't know about th kinetic theory of heat, which eventually supplanted caloric theory. This exercise gave me personally a lot of emp for the calorists of old.

What I find most interesting about caloric theory is that although it was wrong, *it yielded insights which we still true today*.

One notable example is the Carnot cycle. Quoting wikipedia:

> *Sadi Carnot, who reasoned purely on the basis of the caloric theory, developed his principle of the Carnot cyc which still forms the basis of heat engine theory. Carnot's analysis of energy flow in steam engines (1824) m the beginning of ideas which led thirty years later to the recognition of the second law of thermodynamics.*

In other words, the road to the heat engine theory and eventually the second law of thermodynamics was pave part, by an incorrect theory of heat. Another success of caloric theory was a correction to Newton's calculation speed of sound in air, which held for nearly a century afterward.

Implications for Interpretability

I think there are many lessons we as interpretability researchers can learn from the history of caloric theory. Ou theories will probably be wrong, and we should be willing to change our theories in the face of experimental evidence. Designing experiments which demonstrate that those theories are wrong will be a central challenge f But the more subtle point that I want to emphasize is that *wrong theories can still provide real utility*. Even if we the superposition hypothesis will be disproven in the future, which it may very well be, using it is not a fool's er There is still hope that it will be "correct enough" to illuminate practical safety wins and even scientific understa which outlive the superposition hypothesis itself.

Open Problem: Attribution Dictionary Learning

Chris Olah, Adly Templeton, Trenton Bricken, Adam Jermyn

Ordinary dictionary learning only considers activations. It ignores gradients and weights. It seems like we shoul
able to make it much more efficient if we didn't do this.

More fundamentally, it seems like features have a dual nature. Looking backwards towards the input, they are "representations". Looking forwards towards the output, they are "actions". Both of these should be sparse – th they should sparsely represent the activations produced by the input, and also sparsely affect the gradients influencing the output. Ultimately, it seems like they should be a kind of conjunction of these two kinds of spar

A Concrete Proposal

One operationalization of this might be to change the dictionary learning to ask that the linear attribution be sp

Consider a dictionary learning problem $x \simeq x' = Dy$ where x is the dense activations, y the sparse feature activations, and D the dictionary. A traditional SAE loss might be:

$$L_{SAE} = \underbrace{||x - x'||^2}_{\text{Reconstruction error}} + \underbrace{\lambda||y||_1}_{\text{Activation sparsity penalty}}$$

Recall that the linear attribution vector $A_x = x \odot \nabla_x L_{LLM}$ gives an approximation of how each element of x affects the LLM loss. An analogous attribution vector can be computed for y, $A_y = y \odot \nabla_y L_{LLM}$.[1]

We can then add terms to the SAE loss to encourage this to be sparse and to fully explain the attribution:

$$\begin{aligned} L_{SAE} &= \underbrace{||x - x'||^2}_{\text{Reconstruction error}} + \underbrace{\lambda||y||_1}_{\text{Activation sparsity penalty}} + \underbrace{\alpha||A_y||_1}_{\text{Attribution sparsity penalty}} + \underbrace{\beta\left(\sum A_{x-x'}\right)}_{\text{Unexplained attributior penalty}} \\ &= ||x - x'||^2 + \lambda||y||_1 + \alpha||y \odot \nabla_y L_{LLM}||_1 + \beta|(x-x') \cdot \nabla_x L_{LLM}| \end{aligned}$$

This directly optimizes the sparsity of the attribution vector we recently used to study feature circuits in Using Features For Easy Circuit Identification (see also attribution in Marks *et al.* on circuits of features, Kramár *et al.* o attribution patching, and Olah *et al.* on attribution to neurons in vision models).

We briefly investigated the features produced by this loss in a one-layer transformer. At first glance, they seeme about equally good to our normal features in that context. But we don't consider this at all dispositive. We plan revisit this at some point in the future, but it may not be for a few months, and could be an interesting subject f someone else to investigate in the interim.

Isolating Circuits Paths of Different Lengths

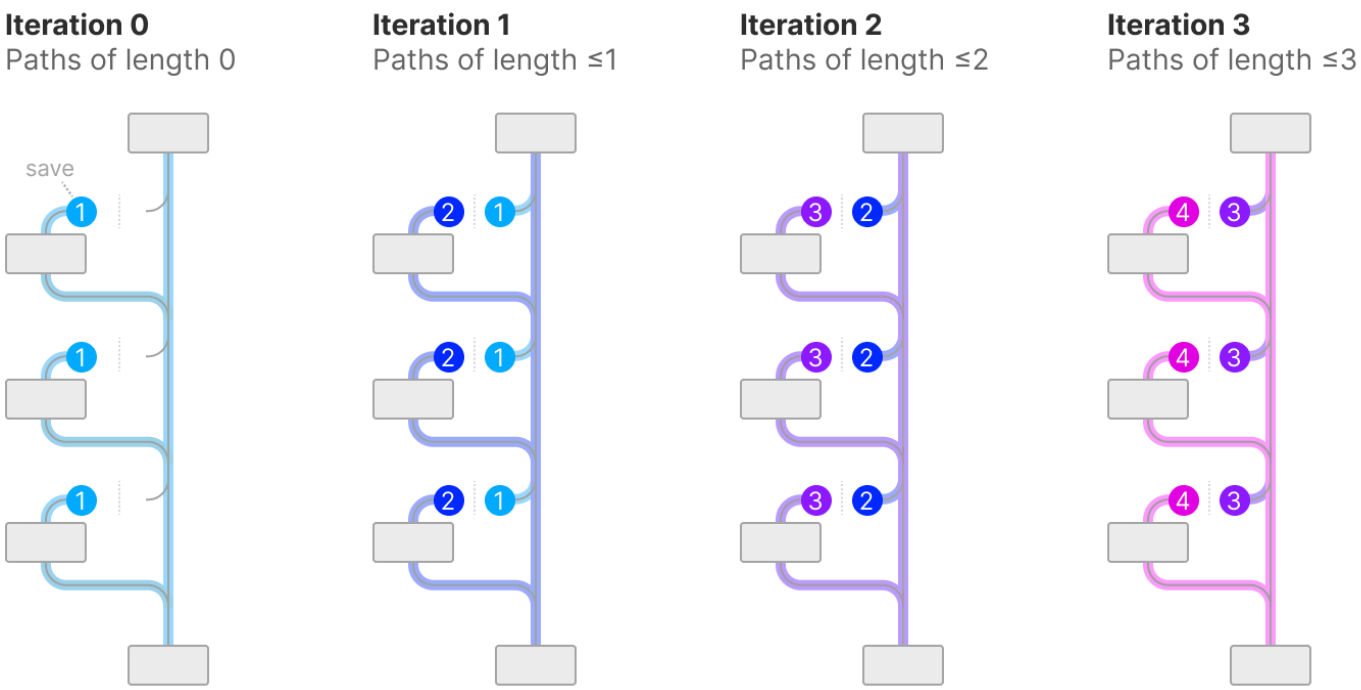
Chris Olah

In *A Mathematical Framework for Transformer Circuits*, we briefly described an algorithm for studying paths of l
at most k:

> **ALGORITHM FOR MEASURING MARGINAL LOSS REDUCTION OF NTH ORDER TERMS**
>
> **Step 1:** Run model, save all attention patterns.
>
> **Step 2:** Run model, forcing all attention patterns to be the version you recorded, and instead of adding attention head outputs to residual stream, save the output, and then replace it with a ze tensor of the same shape. Record resulting loss.
>
> **Step n:** Run model, forcing all attention patterns to be the version you recorded, and instead of adding attention head outputs to residual stream, save the output, and replace it with the value you saved for this head last time. Record resulting loss.

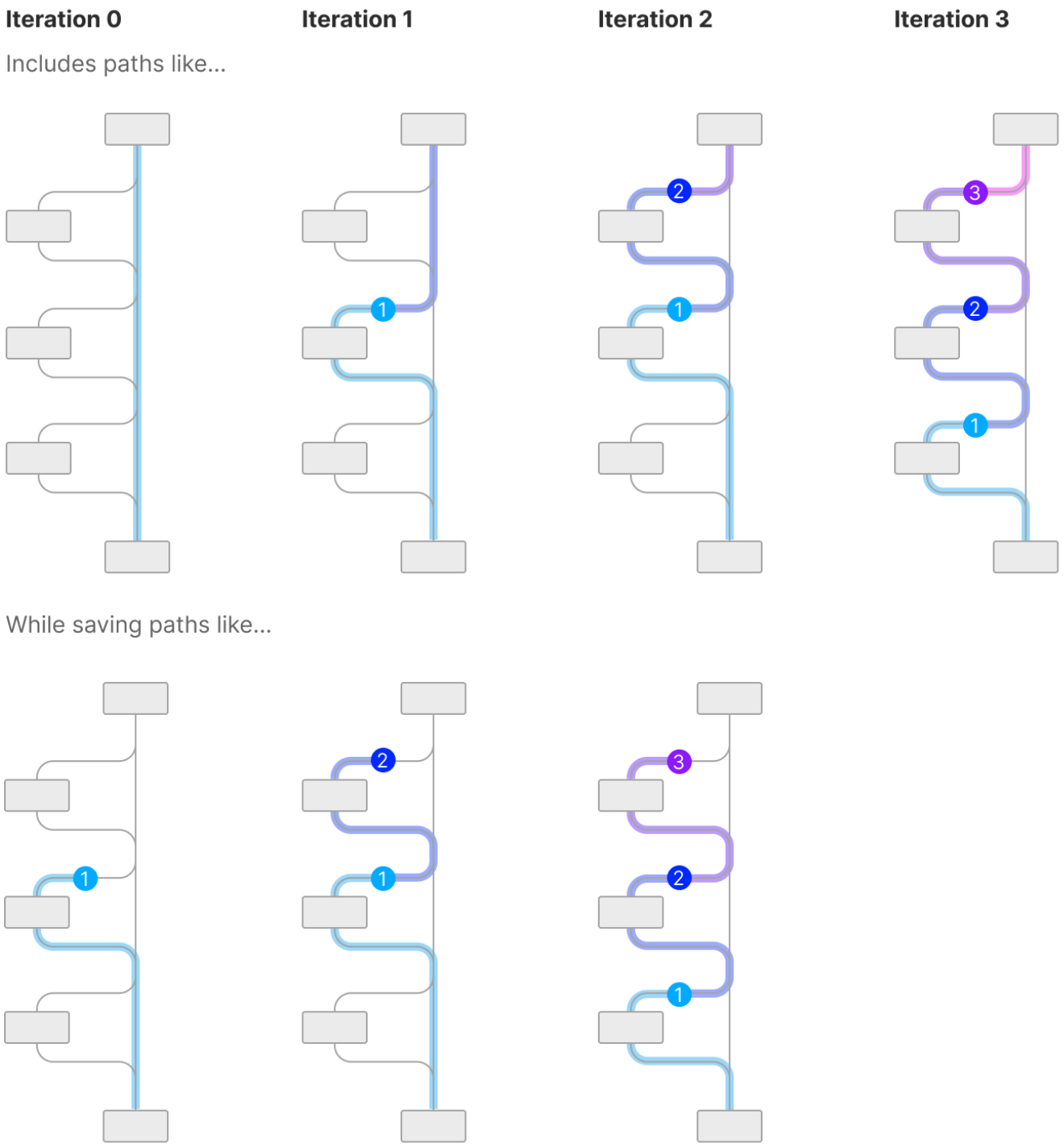
However, we didn't really explain why this algorithm works, and it was easy to miss. We think there are some ca where this algorithm can be interesting, and this update provides a more intuitive explanation.

Let's picture what happens as we apply this algorithm:

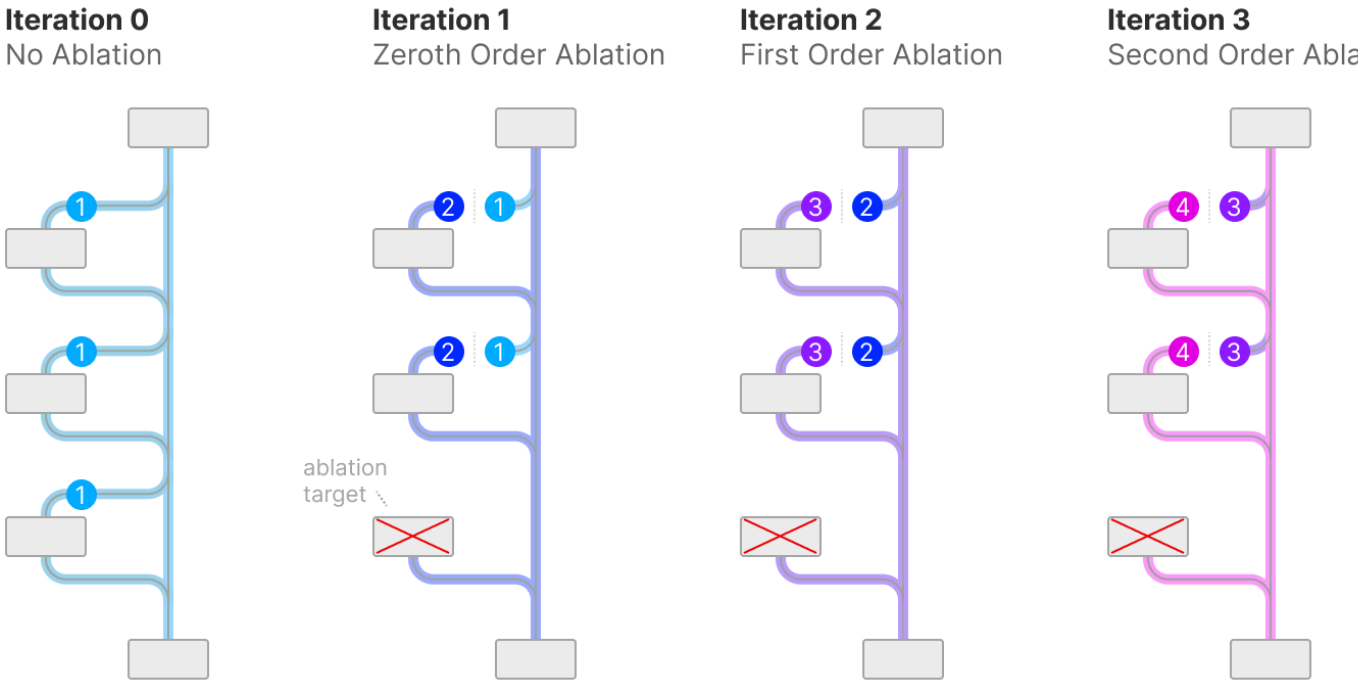


It's easy to see that on the first iteration, we isolate paths of length 0 (ie. we only use the direct path on the resi stream). But also note that we're saving the paths of length 1.

On the next iteration, we use the paths of length 1, and save the paths of length 2. And so on.



This basic idea can be used to create algorithms for isolating all kinds of effects. In particular, modifying the bas case can be very powerful. For example, the following variant can isolate the k-step effects of an ablation:



Other interesting variants are:

- Ablate a component for steps i through j in order to then get the ablations paths greater than one len but less than others.
- For the basic algorithm, feed in the difference between values saved at different stages in order to isola longer paths.

Research By Other Groups

Emmanuel Ameisen, Joshua Batson, Hoagy Cunningham, Anna Golubeva, Jack Lindsey

Finally, we'd like to highlight a selection of recent work by a number of researchers at other groups which we belie be of interest to you if you find our papers interesting. Given the increasing rate of progress in the field, this is far comprehensive, and rather represents a selection of potential interest.

SPARSE FEATURE CIRCUITS

In "Sparse Feature Circuits: Discovering and Editing Interpretable Causal Graphs in Language Models", Marks, R Michaud, Belinkov, Bau, and Mueller investigate scalable methods for identifying model circuits across a range language tasks in a 6L 70M parameter Pythia model. Their main contributions are using dictionary learning wit reconstruction error routed back into the model for attribution, a zero-ablation faithfulness metric and corresponding procedure for unsupervised circuit discovery, a demonstration of editing circuits for model debi and some notes on scalable causal effect estimation.

They train sparse autoencoders (SAEs) separately on residual streams, MLP outputs, and attention outputs, and those feature activations as nodes for the purpose of circuit analysis. They also make nodes for the residuals of autoencoder – this has the advantage of making feature-based decomposition of the model complete, and the disadvantage that the residuals are high-dimensional and hard to interpret. (For example, a zero-feature SAE w present a sparse circuit from this perspective, with every node a residual node, but not be very interpretable.) T then conduct attribution analysis to identify which nodes are most important for a specific model output. Given importance threshold for nodes (features + errors) and edges, they can extract a candidate circuit for a given behavior.

To measure circuit quality, they evaluate interpretability with human raters, who find the sparse features selecte be significantly more interpretable than neurons. They measure faithfulness as the proportion of the model's performance explained by the circuit relative to a mean-ablated model. Completeness is assessed by ablating t circuit and measuring the impact on model performance, showing that removing a small number of nodes fro feature circuits can eliminate the model's task performance.

Editing (ablating) circuits leads to successful model debiasing, which can be improved with post-editing finetun They demonstrate this with Spurious Human-interpretable Feature Trimming (SHIFT), where a human reduces spurious bias from a classifier by editing its feature circuit. The method removes the classifier's dependence on unintended signals identified through circuit analysis while preserving performance on the downstream task. S human judgment is necessary for debiasing – statistics present in the training data for the task may have confo we do not wish to be used, and we have to say what those are. Here, that judgment is expressed in the manual selection of features rather than in a human description (such as prompting with "it is really really important to that race, gender, age, and other demographic characteristics do not influence this decision") or in the curatio gold-standard unbiased dataset (which may be impossible depending on the structure of confounding in the a world). This provides a promising additional method for removing the influence of certain high-level features f model's predictions.

This paper makes some inroads on the important problem of scalable circuit discovery. The reported faithfulnes numbers leave room for future work on identifying additional pathways that contribute to specific model behav even when accounting for SAE reconstruction errors. It is also possible that the large number of nodes necessar recover good performance reveals a problem with current SAEs; while they provide sparse representations of activations, they may not be the sparsest way to represent computation. (For example, there are many "she"- or "her"- in-context features in early layers of one circuit which might be merged in a more compositional representation.) It will be exciting to see these (and related) methods applied to larger models on more comple tasks.

"PATHOLOGICAL" SAE RECONSTRUCTION ERRORS

In "SAE reconstruction errors are (empirically) pathological", Gurnee examines how well sparse autoencoder (SA reconstructions obtained with the standard training objective faithfully preserve the model's next-token predict and in particular whether SAE reconstruction errors are more impactful than similarly sized deviations from the activations.

Gurnee conducts a series of experiments on GPT2-small, comparing the impact of substituting the original activ x with either their SAE reconstruction SAE(x), and the impact of substituting x with a variety of random perturba controls (that match the norm of the perturbation to ||x - SAE(x)||, or the angle of the perturbed vector to the ar between x and SAE(x)). Across all layers of the model, substituting SAE(x) increases the KL divergence between t original and substituted next-token probabilities significantly more than the random perturbations, even contro for differences in norm between x and SAE(x). This gap suggests that SAE reconstructions introduce systematic, than random, errors. Understanding the nature of these errors could help diagnose the shortcomings of SAE reconstructions.

Why might SAEs make systematic errors in reconstruction? There are several possible explanations

- The SAE has not learned all features necessary for a faithful reconstruction of model activity
- The SAE has learned all the features, but the encoder fails to infer the correct feature activations
- Some component of the model activity is not well-described by sparsely active linearly encoded feature is still important for the model's function

Our additional analyses described in this update shed some more light on this finding.

FINE-TUNING

In "A Mechanistic Understanding of Alignment Algorithms: A Case Study on DPO and Toxicity", Lee et al. investi how a fine-tuning algorithm (DPO) changes the behavior of a pretrained model (GPT-2 Medium).

First, the authors identify model components that contribute to toxic behavior by training a probe for toxicity o final layer residual stream, based on a curated dataset of toxic and non-toxic comments. They identify a numbe MLP neurons whose output vectors have high cosine similarity with the probe vector – we may call them 'toxic' neurons – and which produce profane, scatalogical, or insulting tokens when inspected via the logit lens. The to of completions can be reduced by steering, either with the probe vector, the output vector of the most toxic ne or the top principal component of the set of toxic neurons.

In contrast, DPO finetuning optimizes a two-part loss, one which favors non-toxic completions relative to toxic completions of a set of prompts, and the other of which preserves the model output distribution (in the KL sens The resulting finetuned model has a greater reduction in toxic completions than any of the steering approaches provides while, strikingly, every parameter vector in the finetuned model has a 0.99 cosine similarity or above w the original model. "Though its parameters have barely moved,...we show that their collective movement is eno to avoid toxic outputs." This presents a mystery – how can such tiny weight changes produce a substantive cha behavior?

The authors show that many MLP neuron output vectors in the layers preceding the toxic neurons have small changes that decrease the pre-activation of the toxic neurons. (Because GPT2 uses a GeLU activation function, v is slightly negative when inactive, the weights actually shift by a vector positively aligned with the in-weights of toxic neurons.) The sum of those small shifts over many neurons in many layers is enough to substantially chan activation of the toxic neurons, and therefore reduce toxic outputs. One consequence of this infinitesimal appro that the neurons implementing the toxic behavior are still there, and the behavior will resume if they are activat Indeed, the authors show that upscaling the in-weights to the toxic neurons is enough to reactivate the toxic behavior on the same distribution. This speaks to the broader literature on jailbreaks of HHH models; jailbreaks because finetuning reduces the incidence of behavior but doesn't remove its mechanism.

Unalignment via increasing the input weights of the toxic neurons.

BINDING ENTITIES TO ATTRIBUTES

In the paper How do Language Models Bind Entities in Context? (from late 2023) Jiahai Feng and Jacob Steinha investigate the mechanisms by which language models retrieve attributes of entities, studying simple retrieval problems such as “Context: Alice lives in the capital city of France. Bob lives in the capital city of Thailand. Ques Which city does Bob live in?”. In their examples, the attributes and entities are always single tokens.

They find that the information needed to retrieve this connection between entity and attribute seems to be sto entirely in the residual stream of the entity token, so of the tokens “France” and “Thailand” in the previous paragraph’s examples. Given this, with an example of the form “A is X, B is Y”, they need only exchange the resi stream activations of tokens X and Y to make the model ‘believe’ that “A is Y”.

Noting that the assumption that information in activation vectors is stored linearly has been successful in the p they hypothesize that the binding of the attribute to the entity is represented by a sum of an entity vector and “binding vector” which represents some kind of abstract identifier given to the object in this particular context. look at the activations for a given entity in two different retrieval contexts and subtract one from the other to eliminate the common entity vector, thereby isolating the binding vector. Performing this multiple times finds a continuous subspace of valid binding vectors, with this subspace having at least two dimensions, and where the correct answering of the question depends upon the two binding vectors being used to track the two entities b sufficiently different. (It would be interesting to see whether the number of sufficiently-different valid binding directions sets the upper limit on the number entities that the model can simultaneously track, and how this ch with model size).

Perhaps most interestingly, they find that they can transfer activations across different tasks, so if we have “A is and in a separate context we have “B is X”, the residual streams of X and Y can be exchanged to make the mode believe that “A is X” or that “B is Y”, even if A and B denote objects in totally different categories or tasks. This is evidence that when we say “Alice is X”, in the activations above X we’re not associating X with Alice in particular with a more generic entity marker such as “the first named entity”. They also find that the ability to transfer acro tasks in this way grows with model size, up to LLaMA30B. This use of abstract entity trackers is consistent with c unpublished finding that there exist neurons which fire on (among other things) mentions of the first named en a piece of text, regardless of the name (original work by Catherine Olsson).

Footnotes

1. Note that the gradient of y with respect to the LLM loss is easily computed if we save the gradient of the loss with respect to the input acti since $\nabla_y L_{LLM} = D^T \nabla_x L_{LLM}$ where D is the dictionary. [↩]